Best Time to Buy and Sell (easy)

Problem Statement

You are given an array `prices` where `prices[i]` is the price of a given stock on the ith day.

You want to maximize your profit by choosing a **single day** to buy one stock and choosing a **different day in the future** to sell that stock.

Return the maximum profit you can achieve from this transaction. If you cannot achieve any profit, return 0.

Examples

1.
   - **Input:** [3, 2, 6, 5, 0, 3]
   - **Expected Output:** 4
   - **Justification:** Buy the stock on day 2 (price = 2) and sell it on day 3 (price = 6). Profit = 6 - 2 = 4.

2.
   - **Input:** [8, 6, 5, 2, 1]
   - **Expected Output:** 0

- **Justification:** Prices are continuously dropping, so no profit can be made.

3.
   - **Input:** [1, 2]
   - **Expected Output:** 1
   - **Justification:** Buy on day 1 (price = 1) and sell on day 2 (price = 2). Profit = 2 - 1 = 1.

Try it yourself

Try solving this question here:

Java

```java
public class Solution {
    public int maxProfit(int[] prices) {
        // ToDo: Write Your Code Here.
        return 0;
    }
}
```

```
7
```

Test Case | Run Code Result | Stdout

```
[3, 2, 6, 5, 0, 3]
```

Use Example Testcases | Run | Submit

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Solution: Best Time To Buy And Sell Stock

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